

Research Notes

1 Introduction

Document will outline my approach to learning hydraulics and how to create a hydraulic launcher. Hydraulic launchers are interesting because they are used often in rollercoaster launch systems, see figure 1, military drones, and aircrafts. Often, large drones are catapulted with hydraulic launchers because of their ability to deliver high force, and precise speed in a compact area.



Figure 1: *Wicked* rollercoaster from theme park in Utah [1].

I'll detail in this report the exact mechanics of hydraulics and how one can fabricate one from scratch.

2 Hydraulics

Hydraulics move heavy machinery, like construction equipment, around using trapped liquid. That might be ambiguous, but that trapped liquid compresses and exerts massive forces due to pressure. A good example is taking an empty syringe, placing one's finger on the end, and then pushing the plunger until it is physically unable to. When applied force on the plunger is released, the plunger springs back to its original position due to pressure. However, if that syringe is filled with water, the whole scenario becomes widely different, leading us into *Pascal's Law*, the direct mathematical foundation for hydraulics. But, first. let's analyze the physical properties of pressure and its relationship to energy.

2.1 Pressure

Pressure is defined as force per unit area, or:

$$P = \frac{F}{A} \quad (1)$$

and has the SI units Pascal, see figure 2, which are Newtons per square meter. It is often more convenient to use pressure rather than force to describe fluid behavior as they are more complicated systems.

For any object sitting on Earth, it experiences the force of gravity, and thus, the normal force exerted by the surface that it is resting on. That force of gravity is an object's weight, and the normal force is the feeling of having weight. However, for non-point masses, different orientations may yield different areas in contact with the surface, which yields different exerted pressures. Consider a 20 kg box with length 5 metres, width 10 meters, and height 2 meters. If you lay the box on its large side, the area in contact is 50 m^2 , and has a pressure computed out to:

$$P = \frac{200 \text{ N}}{50 \text{ m}^2} = 4 \text{ Pa}$$

But, if the box is laid on a more narrow side, the area in contact is 20 m^2 and the pressure then becomes 10 Pa . Intuitively, it makes sense why the pressure would be more. Standing on our feet, the pressure is much more manual, but if you try to balance on one finger, even just for a second, the pain on that hand becomes unbearable. When the area of contact decreases, the pressure increases. This explains why sharper objects cut easier with less amounts of force. When the tip of a blade is sharpened so that the area of contact is as small as possible, it cuts through materials like butter.

For non-solid objects, pressure is considered differently. For liquids, the medium is treated as a continuous distribution of matter, whereas a gas' pressure is considered to be the average pressure from molecular collisions with the wall. Think *Ideal Gas Law*: $PV = nRT$. When the number of moles increases, there are more gas molecules hitting the wall, increasing the pressure

Due to the nature of equation (1), adding another distance multiplier can change the nature of the equation. Let's consider the box example again. Remember that work is defined as the force applied parallel to the displacement of an object, or:

$$W = F \cdot d \quad (2)$$

and that if adding an additional displacement modifier to area makes the unit m^3 , indicative of volume, where:

$$V = A \cdot d$$

then, one can substitute both of these equations into the numerator and denominator respectively for equation (1), so that:

$$P = \frac{F \cdot d}{A \cdot d} = \frac{W}{V} \quad (3)$$

where the displacement vector cancels. Then, by the definition of the third term in equation (3), pressure can be defined as a measure of the energy per unit volume of an object. Since energy is *stored* per unit volume when pressure is applied, that alludes to the fact that there is a kinetic and potential energy associated with fluids.

If pressure is able to be defined by the energy per unit volume of a liquid, then pressure should be able to be defined as kinetic energy per unit volume of a liquid, as well as potential energy per unit volume of a liquid. If a liquid is moving, then the standard kinetic energy equation for moving objects, being:

$$KE = \frac{1}{2}mv^2 \quad (4)$$

can be substituted into the W portion of equation (3), so that:

$$P = \frac{\frac{1}{2}mv^2}{V}$$

Remember that density is described by how much mass fits into some specific volume, or mathematically:

$$\rho = \frac{m}{V} \quad (5)$$

which can then be substituted in the prior equation where:

$$P = \frac{1}{2}\rho v^2 \quad (6)$$

Furthermore, one can define pressure in terms of potential energy per unit volume as well, which can be used by deriving the potential energy for a mass suspended midair. If work equals the force acting parallel to the displacement to an object, as seen in equation (2), and if intuitively, the object does fall some amount of height h , then the potential energy stored in that object is:

$$U = mg \cdot h \quad (7)$$

which can then be substituted into equation (3). The reason this is the case is because, technically, water molecules are subject to gravity and will tend to *fall*, thus, storing energy. With further consideration to the density equation (5), pressure is defined as:

$$P = \rho gh \quad (8)$$

This equation is also known as the hydrostatic pressure of a liquid, which is the pressure exerted by a fluid at rest or in equilibrium, caused by the force of gravity. It is the weight of a liquid pushing down on itself and all other walls that surround it, and explain why it is unsafe to go deep into the ocean because hydrostatic pressure becomes so high. It also explains why one's ears will pop if they dive into water. When going deeper into water, the hydrostatic pressure increases overhead, causing a subsequent equalizing reaction of the ears to balance pressure.

2.2 Pascal's Law



Figure 2: French mathematician, philosopher, physicist, and prose writer; discovered Pascal's Law [2]

When the syringe is instead filled with a liquid such as water, the fluid is not as compressible in comparison to air, and begins to exert its pressure forces on the exterior of the syringe. Pascal's law restates this phenomena, claiming that any pressure applied to a confined liquid exerts forces equally and undiminished (does not weaken) in all directions to every part of the whole fluid and the walls of its container.

References

- [1] Nick Weisenberger. Launch systems. *Coasters-101*, 2013.
- [2] Lucien Jerphagnon. Blaise pascal. *Encyclopedia Britannica — Britannica Encyclopedia Britannica — Britannica Encyclopedia Britannica — Britannica Encyclopedia Britannica*, August 2026.